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6.6 Answer Explanations: Data Sufficiency

The following discussion of Data Insights is intended to familiarize you with the most efficient and effective approaches to the kinds of problems common to Data Insights. The particular questions in this chapter are generally representative of the kinds of Data Insights questions you will encounter on the GMAT exam. Remember that it is the problem-solving strategy that is important, not the specific details of a particular question.

Questions 273 to 319 — Difficulty: Easy

273. What is the number of pages of a certain journal article?

1. The size of each page is $5\frac{1}{2}$ inches by 8 inches.
2. The average (arithmetic mean) number of words per page is 250.

Arithmetic Applied Problems

1. Given that each page is $5\frac{1}{2}$ inches by 8 inches, any positive integer can be the number of pages of the journal article. For example, it is possible that the journal article consists of 10 pages, each $5\frac{1}{2}$ inches by 8 inches. However, it is also possible that the journal article consists of 20 pages, each $5\frac{1}{2}$ inches by 8 inches. Therefore, the number of pages cannot be determined; NOT sufficient.
2. Given that the average number of words per page is 250, any positive integer can be the number of pages of the journal article. For example, it is possible that the journal article consists of 10 pages and contains a total of 10(250) = 2,500 words, and in this case the average number of words per page is $\frac{2{,}500}{10} = 250$. However, it is also possible that the journal article consists of 20 pages and contains a total of 20(250) = 5,000 words, and in this case the average number of words per page is also $\frac{5{,}000}{20} = 250$. Therefore, the number of pages cannot be determined; NOT sufficient.

   Taking (1) and (2) together, it is still not possible to determine the number of pages because each of the examples used above satisfies both (1) and (2).

   The correct answer is E; both statements together are still not sufficient.

274. If a certain vase contains only roses and tulips, how many tulips are there in the vase?

1. The number of roses in the vase is 4 times the number of tulips in the vase.
2. There is a total of 20 flowers in the vase.

Arithmetic Ratio and Proportion

The task is to determine the number of tulips in the vase.

1. This says that for every tulip in the vase, there are 4 roses, so the ratio of roses to tulips is 4:1. Because there could be 4 roses and 1 tulip or there could be 20 roses and 5 tulips, the number of tulips cannot be uniquely determined; NOT sufficient.
2. This says the total number of roses and tulips in the vase is 20. Because there could be 10 roses and 10 tulips or there could be 15 roses and 5 tulips, the number of tulips cannot be uniquely determined; NOT sufficient.

From (1), the ratio of roses to tulips is 4:1 so $\frac{1}{5}$ of the flowers in the vase are tulips. From (2), there are 20 flowers in the vase. So, taking (1) and (2) together, the number of tulips in the vase is $\frac{1}{5}(20) = 4$.

The correct answer is C; both statements together are sufficient.

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275. The cost of 10 pounds of apples and 2 pounds of grapes was $12. What was the cost per pound of apples?

1. The cost per pound of grapes was $2.
2. The cost of 2 pounds of apples was less than the cost of 1 pound of grapes.

Arithmetic Applied Problems

Determine the cost per pound of apples.

1. The cost per pound of grapes was $2. The cost of 2 pounds of grapes was 2($2) = $4. The cost of the 10 pounds of apples was $12 − $4 = $8. The cost per pound of the apples was $\frac{\$8}{10} = \$0.80$; SUFFICIENT.
2. Intuitively, since (2) involves an inequality, it seems as if it wouldn't give enough information to determine the unique cost per pound of apples. A good strategy might be to guess multiple values for the costs per pound of apples and grapes that satisfy the condition given in the problem (the cost of 10 pounds of apples and 2 pounds of grapes is $12) and look for values that also meet the condition in (2), namely that the cost of 2 pounds of apples is less than the cost of 1 pound of grapes. For example:

Cost per pound of apples	Cost per pound of grapes	Total cost of 10 pounds of apples and 2 pounds of grapes, in dollars	Is total cost $12?	Is the cost of 2 pounds of apples less than the cost of 1 pound of grapes?
$1	$1	10(1) + 2(1)	yes	No, 2(1) $\not<$ 1
$0.50	$3.50	10(0.5) + 2(3.5)	yes	Yes, 2(0.50) < 3.50
$0.60	$3.00	10(0.6) + 2(3)	yes	Yes, 2(0.60) < 3.00

As shown in the table, the cost per pound of apples could be $0.50 or $0.60, so the cost per pound of apples cannot be uniquely determined; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

276. What was the median annual salary for the employees at Company X last year?

1. Last year, there were 29 employees at Company X.
2. Last year, 12 employees at Company X had an annual salary of $24,000.

Arithmetic Statistics

1. Given that there were 29 employees, the median salary is the 15th salary in a list of 29 positive numbers, but we cannot determine the value of any of the numbers; NOT sufficient.
2. Given that 12 employees had an annual salary of $24,000, the median salary is the median of a list of 12 or more positive numbers, of which 12 are equal to $24,000. If there were only 12 employees, then the median salary would be $24,000. However, if there were more than 2(12) = 24 employees and all but 12 of the employees had an annual salary of $30,000, then the median salary would be $30,000; NOT sufficient.

Taking (1) and (2) together, if 12 employees had an annual salary of $24,000 and the remaining 17 employees had an annual salary of $20,000, then the median salary would be $20,000. However, if 12 employees had an annual salary of $24,000 and the remaining 17 employees had an annual salary of $30,000, then the median salary would be $30,000.

Tip: If more than half the numbers in a list of numbers have the same value, then the median of the list will be that value, regardless of the values of the remaining numbers.

The correct answer is E; both statements together are still not sufficient.

277. How many basic units of Currency X are equivalent to 250 basic units of Currency Y?

1. 100 basic units of Currency X are equivalent to 625 basic units of Currency Y.
2. 2,000 basic units of Currency X are equivalent to 12,500 basic units of Currency Y.

Arithmetic Ratio and Proportion

1. The equivalency of 100 basic units of Currency X and 625 basic units of Currency Y means that 1 basic unit of Currency Y is equivalent to $\frac{100}{625}$ basic units of Currency X. Therefore, 250 basic units of Currency

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unit of Currency Y is equivalent to $\frac{100}{625}$ basic units of Currency X. Therefore, 250 basic units of Currency Y are equivalent to $(250)\left(\frac{100}{625}\right)$ basic units of Currency X; SUFFICIENT.

2. The equivalency of 2,000 basic units of Currency X and 12,500 basic units of Currency Y means that 1 basic unit of Currency Y is equivalent to $\frac{2{,}000}{12{,}500}$ basic units of Currency X. Therefore, 250 basic units of Currency Y are equivalent to $(250)\left(\frac{2{,}000}{12{,}500}\right)$ basic units of Currency X; SUFFICIENT.

Tip: For Data Insights questions in which the task is to determine a specific value, there is no need to perform calculations to get the value in its simplest form. The goal is to determine whether the value can be uniquely determined, not to find the actual value in its simplest form.

The correct answer is D; each statement alone is sufficient.

278. A company bought 3 printers and 1 scanner. What was the price of the scanner?

1. The total price of the printers and the scanner was $1,300.
2. The price of each printer was 4 times the price of the scanner.

Algebra Simultaneous Equations

Let P_1, P_2, and P_3 be the prices, in dollars, of the 3 printers and let S be the price, in dollars, of the scanner. What is the value of S?

1. Given that $P_1 + P_2 + P_3 + S = 1{,}300$, it is clear that the value of S cannot be determined; NOT sufficient.
2. Given that $P_1 = P_2 = P_3 = 4S$, it is possible that $P_1 = P_2 = P_3 = 400$ and $S = 100$. On the other hand, it is also possible that $P_1 = P_2 = P_3 = 800$ and $S = 200$; NOT sufficient.

Taking (1) and (2) together, substituting $P_1 = 4S$, $P_2 = 4S$, and $P_3 = 4S$ into $P_1 + P_2 + P_3 + S = 1{,}300$ gives $4S + 4S + 4S + S = 1{,}300$, or $13S = 1{,}300$, and therefore $S = 100$.

The correct answer is C; both statements together are sufficient.

279. Each of the 256 solid-colored marbles in a box is either blue, green, or purple. What is the ratio of the number of blue marbles to the number of purple marbles in the box?

1. The number of green marbles in the box is 4 times the number of blue marbles in the box.
2. There are 192 green marbles in the box.

Arithmetic Ratio and Proportion

Let B, G, and P be the numbers of marbles, respectively, that are blue, green, and purple. From the given information, it follows that $B + G + P = 256$. What is the value of $\frac{B}{P}$?

1. Given that $G = 4B$, it is possible that $\frac{B}{P} = \frac{1}{251}$ (choose $B = 1$, $G = 4$, and $P = 251$), and it is possible that $\frac{B}{P} = \frac{1}{123}$ (choose $B = 2$, $G = 8$, and $P = 246$); NOT sufficient.
2. Given that $G = 192$, it is possible that $\frac{B}{P} = \frac{1}{63}$ (choose $B = 1$, $G = 192$, and $P = 63$), and it is possible that $\frac{B}{P} = \frac{1}{31}$ (choose $B = 2$, $G = 192$, and $P = 62$); NOT sufficient.

Taking (1) and (2) together, it follows from $G = 4B$ and $G = 192$ that $192 = 4B$, or $B = 48$. Thus, $B + G + P = 256$ becomes $48 + 192 + P = 256$, and so $P = 256 - 48 - 192 = 16$. Therefore, $\frac{B}{P} = \frac{48}{16}$.

The correct answer is C; both statements together are sufficient.

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The correct answer is C; both statements together are sufficient.

280. A certain mixture of paint requires blue, yellow, and red paints in ratios of 2:3:1, respectively, and no other ingredients. If there are ample quantities of the blue and red paints available, is there enough of the yellow paint available to make the desired amount of the mixture?

1. Exactly 20 quarts of the mixture are needed.
2. Exactly 10 quarts of the yellow paint are available.

Arithmetic Ratios

Given that the mixture requires blue paint, yellow paint, and red paint in the ratios 2:3:1, it follows that $\frac{3}{2+3+1} = \frac{1}{2}$ of the mixture will be yellow paint. Determining whether there is enough yellow paint available depends on how much of the mixture is needed and how much yellow paint is available.

1. This indicates that exactly 20 quarts of the paint mixture are needed, so $\frac{1}{2}(20) = 10$ quarts of yellow paint are needed. However, there is no information about how much yellow paint is available; NOT sufficient.
2. This indicates that exactly 10 quarts of yellow paint are available, but there is no information about how much of the mixture or how much yellow paint is needed; NOT sufficient.

Taking (1) and (2) together, 10 quarts of yellow paint are needed, and 10 quarts are available.

The correct answer is C; both statements together are sufficient.

281. There are 2 groups of students who took a history test. Was the average (arithmetic mean) score of the students in Group A higher than the average score of the students in Group B who took the test?

1. Of the students who took the test, 10 were in Group A and 12 were in Group B.
2. On the test, the highest score was achieved by a Group B student, and the lowest score was achieved by a Group A student.

Arithmetic Statistics

1. Given that 10 of the scores were by students in Group A and 12 of the scores were by students in Group B, the following table shows two possibilities.

A scores	B scores	A average	B average	A average > B average?
10 scores of 8	12 scores of 1	8	1	yes
10 scores of 1	12 scores of 8	1	8	no

Therefore, it is not possible to determine whether the average score of the students in Group A is higher than the average score of the students in Group B; NOT sufficient.

Given that the highest score was achieved by a Group B student and the lowest score was achieved by a Group A student, the following table shows two possibilities.

A scores	B scores	A average	B average	A average > B average?
1, 7, 7	2, 2, 8	5	4	yes
1, 1, 1	8, 8, 8	1	8	no

Therefore, it is not possible to determine whether the average score of the students in Group A is higher than the average score of the students in Group B; NOT sufficient.

Taking (1) and (2) together, the table below shows that it is not possible to determine whether the average score of the students in Group A is higher than the average score of the students in Group B.

A scores		B scores		A average	B average	A average > B average?
score	frequency	score	frequency			
2	1	3	11	4.7	4	yes
5	9	15	1			
score	frequency	score	frequency			

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score	frequency	score	frequency			
2	1	6	9	4.7	6	no
5	9					

The correct answer is E; both statements together are still not sufficient.

282. The research funds of a certain company were divided among three departments, X, Y, and Z. Which one of the three departments received the greatest proportion of the research funds?

1. The research funds received by Departments X and Y were in the ratio 3 to 5, respectively.
2. The research funds received by Departments X and Z were in the ratio 2 to 1, respectively.

Algebra Order; Ratio

Let x, y, and z be the research funds, respectively, of Departments X, Y, and Z. Which of $\frac{x}{x+y+z}$, $\frac{y}{x+y+z}$, or $\frac{z}{x+y+z}$ is the greatest, or respectively, which of x, y, or z is the greatest?

1. Given that $\frac{x}{y} = \frac{3}{5}$ where x and y are positive, it follows that $x < y$. However, nothing is known about z other than z is positive, so it cannot be determined which of x, y, or z is the greatest; NOT sufficient.
2. Given that $\frac{x}{z} = \frac{2}{1}$ where x and z are positive, it follows that $z < x$. However, nothing is known about y other than y is positive, so it cannot be determined which of x, y, or z is the greatest; NOT sufficient.

Taking (1) and (2) together, it follows from (1) that $x < y$ and it follows from (2) that $z < x$. Therefore, $z < x < y$, and y is the greatest.

The correct answer is C; both statements together are sufficient.

283. In a certain class, some students donated cans of food to a local food bank. What was the average (arithmetic mean) number of cans donated per student in the class?

1. The students donated a total of 56 cans of food.
2. The total number of cans donated was 40 greater than the total number of students in the class.

Arithmetic Statistics

Let s be the number of students and let c be the number of cans. What is the value of $\frac{c}{s}$?

1. Given that $c = 56$, it is not possible to determine the value of $\frac{c}{s}$ because nothing is known about the value of s other than s is a positive integer; NOT sufficient.
2. Given that $c = 40 + s$, it is not possible to determine the value of $\frac{c}{s}$. For example, if $c = 40$ and $s = 40$, then $\frac{c}{s} = 1$. However, if $c = 80$ and $s = 40$, then $\frac{c}{s} = 2$; NOT sufficient.

Taking (1) and (2) together, it follows that $40 + s = c = 56$, or $s = 56 - 40 = 16$. Therefore, $c = 56$, $s = 16$ and $\frac{c}{s} = \frac{56}{16}$.

The correct answer is C; both statements together are sufficient.

284. Each of the n employees at a certain company has a different annual salary. What is the median of the annual salaries of the n employees?

1. When the annual salaries of the n employees are listed in increasing order, the median is the 15th salary.
2. The sum of the annual salaries of the n employees is $913,500.

Arithmetic Statistics

1. Given that the median is the 15th salary and all n salaries are different, it follows that there are 14 salaries

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1. Given that the median is the 15th salary and all n salaries are different, it follows that there are 14 salaries less than the 15th salary and 14 salaries greater than the 15th salary, for a total of 29 employee salaries. However, there is no information about what any of the salaries actually are, so the median of the salaries cannot be determined; NOT sufficient.

2. Given that the sum of the annual salaries is $913,500, the median salary cannot be uniquely determined. To determine more than one possible value for the median salary, use common divisibility rules to find factors of $913,500.

   For example, 913,500 is divisible by 5 because it ends in 0. Then, because 913,500 = 5(182,700), there could be 5 salaries, each $182,700. But the salaries must be all different, so we need 5 different salaries that total $913,500. They could be the salaries shown in the table below. Note that the pattern starts with the 3rd (middle) salary of $182,700, and then the 2nd and 4th salaries are "paired" in that one is decreased and the other is increased by the same amount from $182,700. Likewise, for the 1st and 5th salaries. The median is $182,700.

Salary #	Salary
1	$182,698
2	$182,699
3	**$182,700**
4	$182,701
5	$182,702
Total	$913,500

   Also, 913,500 is divisible by 9 because the sum of its digits is divisible by 9, and 913,500 = 9(101,500). Using ideas similar to those used in the first example, there could be 9 different salaries that total $913,500. Using the same technique as in the first example, these salaries could be those shown below. All 9 salaries are listed to reinforce the technique. They need not be written out as part of the solution.

Salary #	Salary
1	$101,496
2	$101,497
3	$101,498
4	$101,499
5	**$101,500**
6	$101,501
7	$101,502
8	$101,503
9	$101,504
Total	$913,500

   Therefore, the median salary could be $182,700 or $101,500; NOT sufficient.

   Taking (1) and (2) together, there are 29 different salaries and their sum is $913,500, but this information is not sufficient to uniquely determine the median.

Salary number	Salary (Example I)	Salary (Example II)
1	$30,100	$30,100
2	$30,200	$30,200
3	$30,300	$30,300
⋮	⋮	⋮
14	$31,400	$31,380
15	**$31,500**	**$31,540**

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16	$31,600	$31,580
⋮	⋮	⋮
27	$32,700	$32,700
28	$32,800	$32,800
29	$32,900	$32,900
Total	$913,500	$913,500

Note that 913,500 = 29(31,500). The table above shows two examples, each consisting of 29 different salaries (not all are shown, however) that were obtained using techniques similar to those used in the previous examples. Note that, in both examples, the 1st salary is $1,400 less than $31,500 and the 29th is $1,400 more than $31,500, the 2nd salary is $1,300 less than $31,500 and the 28th is $1,300 more than $31,500, and so on. The salary columns in both examples are identical except for the 14th, 15th, and 16th salaries, which were used to obtain different medians.

In each example, the median is the 15th salary, so (1) is satisfied, and the sum of the salaries is $913,500, so (2) is satisfied. But the median in one case is $31,500 and the median in the other case is $31,540.

The correct answer is E; both statements together are still not sufficient.

285. In a recent town election, what was the ratio of the number of votes in favor of a certain proposal to the number of votes against the proposal?

1. There were 60 more votes in favor of the proposal than against the proposal.
2. There were 240 votes in favor of the proposal.

Arithmetic Ratio and Proportion

Let F be the number of votes in favor and let A be the number of votes against. What is the value of $\frac{F}{A}$?

1. Given that $F = 60 + A$, it is possible that $\frac{F}{A} = 2$ (choose $F = 120$ and $A = 60$) and it is possible that $\frac{F}{A} = 3$ (choose $F = 90$ and $A = 30$); NOT sufficient.
2. Given that $F = 240$, it is possible that $\frac{F}{A} = 1$ (choose $F = 240$ and $A = 240$) and it is possible that $\frac{F}{A} = 2$ (choose $F = 240$ and $A = 120$); NOT sufficient.

Taking (1) and (2) together, substitute $F = 240$ into $F = 60 + A$ to get $240 = 60 + A$, or $A = 180$. Therefore, $\frac{F}{A} = \frac{240}{180}$.

The correct answer is C; both statements together are sufficient.

286. How many men are in a certain company's vanpool program?

1. The ratio of men to women in the program is 3 to 2.
2. The men and women in the program fill 6 vans.

Arithmetic Applied Problems

Let m be the number of men in the program and let w be the number of women in the program. What is the value of m?

1. Given that $\frac{m}{w} = \frac{3}{2}$, it is not possible to determine the value of m. For example, if $m = 3$ and $w = 2$, then $\frac{m}{w} = \frac{3}{2}$ is true; and if $m = 6$ and $w = 4$, then $\frac{m}{w} = \frac{3}{2}$ is true; NOT sufficient.
2. Given that the men and women fill 6 vans, it is clearly not possible to determine the value of m; NOT sufficient.

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sufficient.

Taking (1) and (2) together, it is still not possible to determine the value of m. Even if it is assumed that each van has the same maximum capacity, it is possible that $m = 18$ and $w = 12$ (a total of 30 people, which would fill 6 vans each having a capacity of 5 people), and it is possible that $m = 36$ and $w = 24$ (a total of 60 people, which would fill 6 vans each having a capacity of 10 people).

The correct answer is E; both statements together are still not sufficient.

287. Last semester, Professor K taught two classes, A and B. Each student in Class A handed in 7 assignments, and each student in Class B handed in 5 assignments. How many students were in Class A?

1. The students in both classes combined handed in a total of 85 assignments.
2. There were 10 students in Class B.

Algebra Simultaneous Equations

Let a be the number of students in Class A and let b be the number of students in Class B. Then the students in Class A handed in a total of $7a$ assignments, and the students in Class B handed in a total of $5b$ assignments. What is the value of a?

1. Given that $7a + 5b = 85$, it is not possible to determine the value of a. For example, it is possible that $a = 5$ and $b = 10$, since 7(5) + 5(10) = 85. On the other hand, it is also possible that $a = 10$ and $b = 3$, since 7(10) + 5(3) = 85; NOT sufficient.
2. Given that $b = 10$, it is not possible to determine the value of a. For example, it is possible that $a = b = 10$; and it is also possible that $a = 5$ and $b = 10$; NOT sufficient.

Taking (1) and (2) together, substituting $b = 10$ into $7a + 5b = 85$ gives $7a + 5(10) = 85$, or $7a = 35$, or $a = 5$.

The correct answer is C; both statements together are sufficient.

288. Was the amount of John's heating bill for February greater than it was for January?

1. The ratio of the amount of John's heating bill for February to that for January was $\frac{26}{25}$.
2. The sum of the amounts of John's heating bills for January and February was $183.60.

Arithmetic Applied Problems

Let J and F be the amounts, respectively and in dollars, of the heating bills for January and February. Is $F > J$?

1. Given that $\frac{F}{J} = \frac{26}{25}$, it follows that $\frac{F}{J} > 1$. Multiplying both sides of the inequality $\frac{F}{J} > 1$ by the positive quantity J gives $F > J$; SUFFICIENT.
2. Given that $J + F = 183.60$, it is not possible to determine whether $F > J$. For example, if $J = 83.60$ and $F = 100.00$, then $J + F = 183.60$ and $F > J$. On the other hand, if $J = 100.00$ and $F = 83.60$, then $J + F = 183.60$ and $F < J$; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

289. Machine R and Machine S work at their respective constant rates. How much time does it take Machine R, working alone, to complete a certain job?

1. The amount of time that it takes Machine S, working alone, to complete the job is $\frac{3}{4}$ the amount of time that it takes Machine R, working alone, to complete the job.
2. Machine R and Machine S, working together, take 12 minutes to complete the job.

Algebra Applied Problems

1. Given that Machine S takes $\frac{3}{4}$ as long as Machine R to complete the job, if the rate of each machine were doubled, then Machine S would still take $\frac{3}{4}$ as long as Machine R to complete the job and Machine R,

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working alone, would then take half as long to complete the job as compared to before the rates were doubled; NOT sufficient.

2. Given that Machine R and Machine S, working together, take 12 minutes to complete the job, if Machine S is extremely slow and thus contributes very little when the machines are working together, then it would take Machine R, working alone, only a little more than 12 minutes to complete the job. However, if Machine S, working alone, takes only a few seconds less than 12 minutes to complete the job, and thus Machine R contributes very little when the machines are working together, then it would take Machine R, working alone, a long time to complete the job; NOT sufficient.

Taking (1) and (2) together, let t be the time it takes for Machine R, working alone, to complete the job. Then the time it takes Machine S, working alone, to complete the job is $\frac{3}{4}t$. Thus, the individual constant rates of Machines R and S are, respectively $\frac{1}{t}$, and $\frac{1}{\frac{3}{4}} = \left(\frac{4}{3}\right)\frac{1}{t}$, and their combined rate is $\left(1+\frac{4}{3}\right)\frac{1}{t} = \frac{7}{3t}$.

Since it takes 12 minutes to complete the job when the machines are working together, it follows that $\left(\frac{7}{3t}\right)\left(12\right) = 1$, which can be solved for t.

The correct answer is C; both statements together are sufficient.

290. Beth's bank charges a service fee on a regular checking account for each month in which the balance on the account falls below $100 at any time during the month. Did the bank charge a service fee on Beth's regular checking account last month?

1. During last month, a total of $1,000 was withdrawn from Beth's regular checking account.
2. At the beginning of last month, Beth's regular checking account balance was $500.

Arithmetic Equations

1. Given that a total of $1,000 was withdrawn from the account last month, it is possible that the bank charged a service fee (for example, if the account had $1,050 at the beginning of the month and no deposits were made), and it is possible that the bank did not charge a service fee (for example, if the account had $1,500 at the beginning of the month); NOT sufficient.
2. Given that the account had $500 at the beginning of last month, it is possible that the bank charged a service fee (for example, if a withdrawal of $450 was made and no deposits were made), and it is possible that the bank did not charge a service fee (for example, if no withdrawals were made); NOT sufficient.

Taking (1) and (2) together, it is still not possible to determine whether the bank charged a service fee last month. For example, if the account had $500 at the beginning of last month and the only activity was a deposit of $1,000 followed later by a withdrawal of $1,000, then the bank would not have charged a service fee. On the other hand, if the account had $500 at the beginning of last month and the only activity was a deposit of $550 followed by a withdrawal of $1,000, then the bank would have charged a service fee.

The correct answer is E; both statements together are still not sufficient.

291. Three houses are being sold through a real estate agent. What is the asking price for the house with the second-largest asking price?

1. The difference between the greatest and the least asking price is $130,000.
2. The difference between the two greater asking prices is $85,000.

Algebra Simultaneous Equations

Let x, y, and z, where $x \leq y \leq z$, be the asking prices of the three houses. This problem can be solved by determining the value of y.

1. This indicates that $z - x = \$130{,}000$, but it does not give the value of y; NOT sufficient.
2. This indicates that $z - y = \$85{,}000$, but it does not give the value of y; NOT sufficient.

Taking (1) and (2) together, x, y, and z could be $100,000, $145,000, and $230,000, respectively, in which case the second-largest asking price is $145,000, or they could be $200,000, $245,000, and $330,000, respectively, in

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the second-largest asking price is $145,000, or they could be $200,000, $245,000, and $330,000, respectively, in which case the second-largest asking price is $245,000.

The correct answer is E; both statements together are still not sufficient.

292. How many people in a group of 50 own neither a fax machine nor a laser printer?

1. The total number of people in the group who own a fax machine or a laser printer or both is less than 50.
2. The total number of people in the group who own both a fax machine and a laser printer is 15.

Algebra Sets

1. This indicates that the total number who own either a fax machine or a laser printer or both is less than 50, but it does not indicate how much less than 50 the total number is. Thus, the number of people in the group who own neither a fax machine nor a laser printer cannot be determined; NOT sufficient.
2. This indicates that, of the 50 people, 15 own both a fax machine and a laser printer, but it does not indicate how many own one or the other. Thus, the number of people in the group who own neither a fax machine nor a laser printer cannot be determined; NOT sufficient.

Taking (1) and (2) together, it is known that 15 people own both a fax machine and a laser printer and that the total number who own either a fax machine or a laser printer or both is less than 50, but the exact number who own neither still cannot be determined. For example, if 20 people own only a fax machine and 10 people own only a laser printer, then both (1) and (2) are true and the number of people who own neither a fax machine nor a laser printer is 50 – 20 – 10 – 15 = 5. However, if 10 people own only a fax machine and 10 people own only a laser printer, then both (1) and (2) are true and the number of people who own neither a fax machine nor a laser printer is 50 – 10 – 10 – 15 = 15.

The correct answer is E; both statements together are still not sufficient.

293. A state legislature had a total of 96 members. The members who did not vote on a certain bill consisted of 25 who were absent and 3 who abstained. How many of those voting voted for the bill?

1. Exactly $\frac{1}{3}$ of the total membership of the legislature voted against the bill.
2. The number of legislators who voted for the bill was 8 more than the total number who were absent or abstained.

Arithmetic Equations

Since there are 96 members and 3 + 25 = 28 of the members did not vote, it follows that 96 – 28 = 68 of the members voted. We are to determine the number of members who voted FOR the bill.

1. We are given that $\frac{1}{3}\left(96\right) = 32$ members voted AGAINST the bill. Therefore, 68 – 32 = 36 members voted FOR the bill; SUFFICIENT.
2. We are given that 8 + (25 + 3) = 36 members voted FOR the bill; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

294. What is the average (arithmetic mean) of the first 90 measurements in a certain list of 92 measurements?

1. The average of the 92 measurements is 7.0 centimeters.
2. The average of the last 2 measurements is 7.2 centimeters.

Arithmetic Statistics

1. Given that the average of all 92 measurements is 7, it follows that the sum of all 92 measurements is 92 × 7 = 644. However, the sum of the first 90 measurements could be any number and hence we cannot determine the average of the first 90 measurements. For example, it is possible that the average of the first 90 measurements is 1 (sum of first 90 measurements is 90 and each of the last 2 measurements is 277), and it is possible that the average of the first 90 measurements is 7 (sum of first 90 measurements is 630 and each of the last 2 measurements is 7); NOT sufficient.
2. Given that the average of the last 2 measurements is 7.2, it follows that the sum of the last 2 measurements is 2 × 7.2 = 14.4. However, the sum of the first 90 measurements, and hence the average of the first 90 measurements, could be any number, and hence we cannot determine the average of the first 90

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measurements; NOT sufficient.

Taking (1) and (2) together, the sum of the first 90 measurements is the sum of all 92 measurements minus the sum of the last 2 measurements, or 644 – 14.4 = 629.6. Therefore, the average of the first 90 measurements is 629.6 ÷ 90.

The correct answer is C; both statements together are sufficient.

295. How many of the employees of a certain company use both a tablet and an iPhone?

1. 60 of the employees use a tablet.
2. 45 of the employees use an iPhone.

Arithmetic Statistics

1. Given that 60 of the employees use a tablet, the number of employees who use both a tablet and an iPhone cannot be determined because no information about iPhone use by employees is provided; NOT sufficient.
2. Given that 45 of the employees use an iPhone, the number of employees who use both a tablet and an iPhone cannot be determined because no information about tablet use by employees is provided; NOT sufficient.

Taking (1) and (2) together, it is still not possible to determine the number of employees who use both a tablet and an iPhone. Indeed, the number of employees who use both could be any integer from 0 to 45, inclusive.

The correct answer is E; both statements together are still not sufficient.

296. What is the standard deviation of the weights of the 30 samples in a certain experiment?

1. The total weight of the 30 samples is 360 grams.
2. Each of the 30 samples weighs 12 grams.

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